PSYC1071: Introduction to Psychology 1
Animal Learning and Cognition

Lecture 3:
Animal Intelligence

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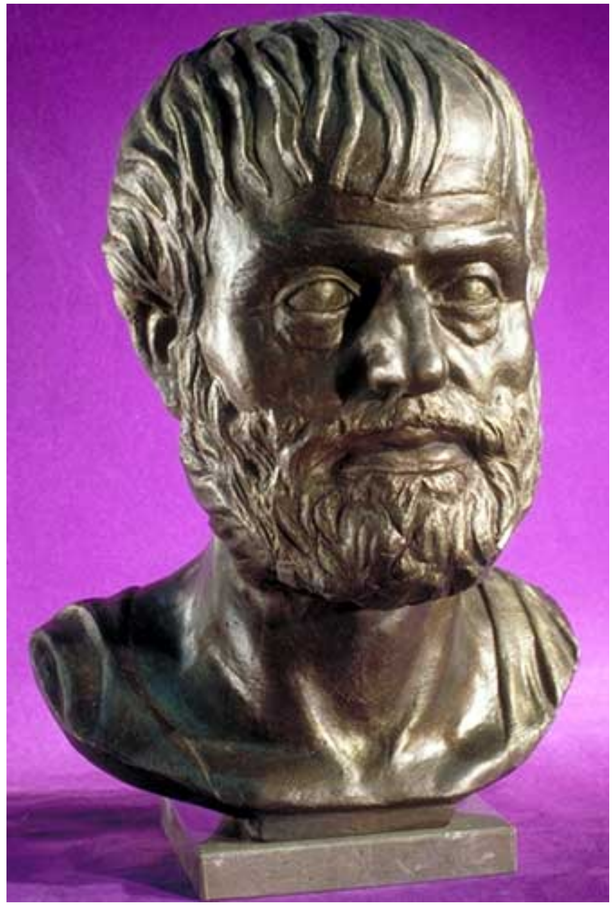
Ranking 'animal intelligence'

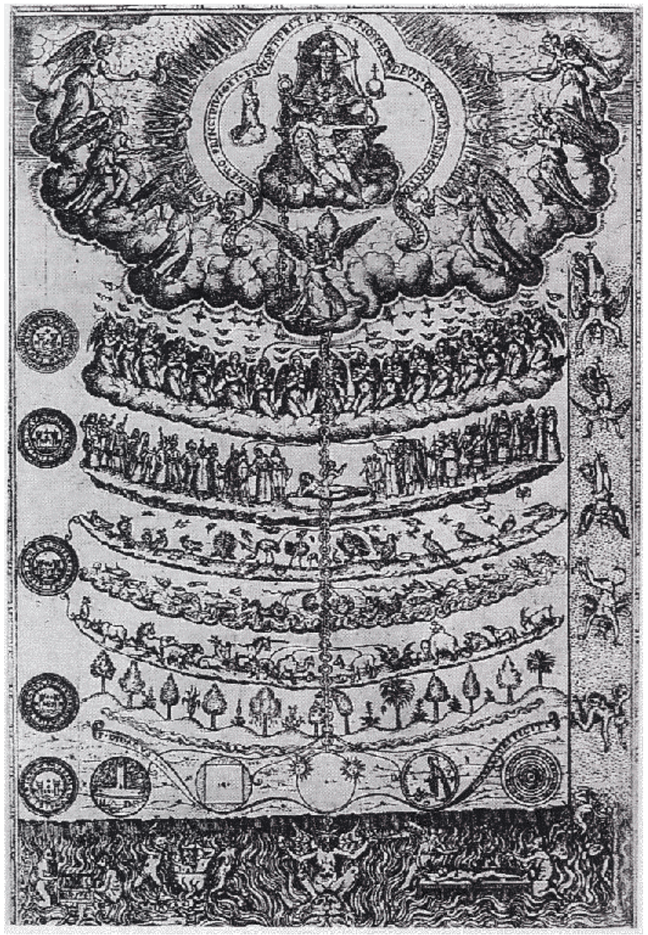
TABLE 1.1 The rank order of the 56 animals students were asked by Nakajima et al. (2002) to score on intelligence, together with the means score assigned to each species. The maximum score was 100

Animal	Score	Animal	Score	Animal	Score	Animal	Score
Chimpanzee	77	Kangaroo	50	Penguin	40	Tuna	25
Orangutan	72	Panda	48	Rabbit	39	Octopus	24
Dolphin	72	Hawk	48	Ostrich	38	Lizard	24
Gorilla	68	Parrot	48	Crocodile	37	Ant	23
Dog	61	Sea lion	48	Rat	35	Frog	23
Baboon	60	Crow	46	Pigeon	33	Carp	22
Whale	57	Cow	45	Sparrow	32	Crab	20
Wolf	56	Giraffe	44	Quail	31	Cockroach	19
Cat	55	Owl	44	Fowl	31	Goldfish	19
Lion	54	Shark	43	Mole	31	Butterfly	17
Bear	52	Sheep	42	Snake	30	Jellyfish	15
Horse	52	Bat	41	Salmon	29	Earthworm	10
Fox	51	Koala	41	Turtle	28	Slug	10
Elephant	50	Pig	41	Honeybee	28	Ameba	8

From Pearce (2008)

Aristotle's Great Chain of Being





Evolution

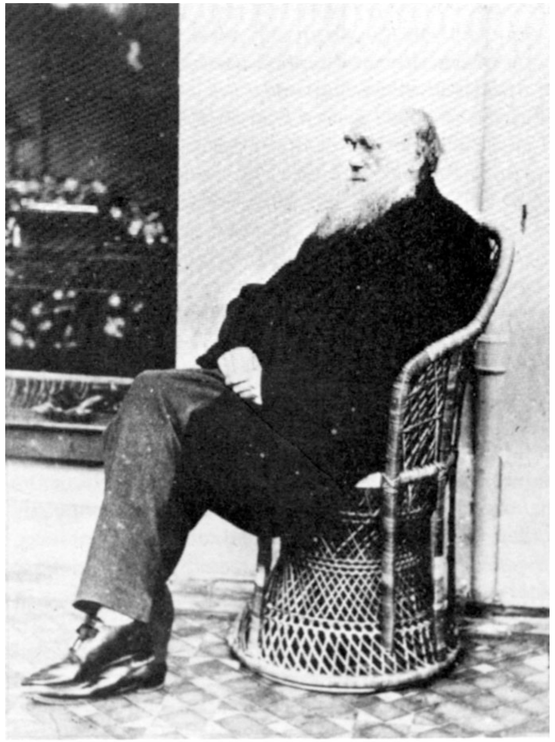
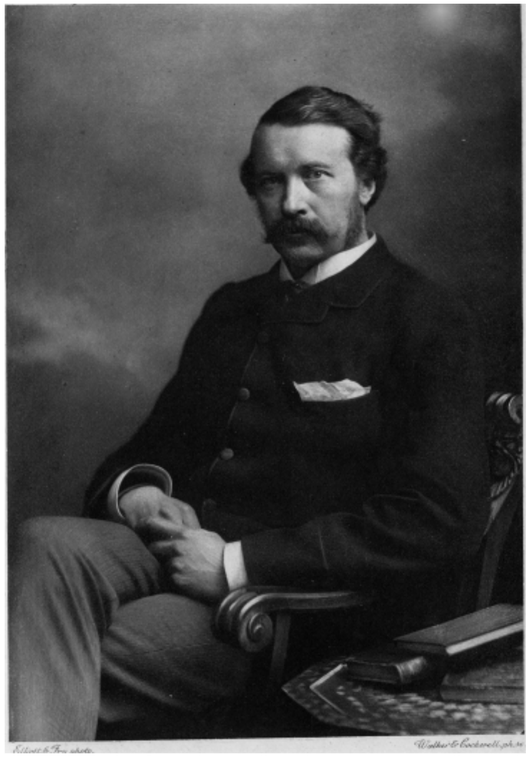
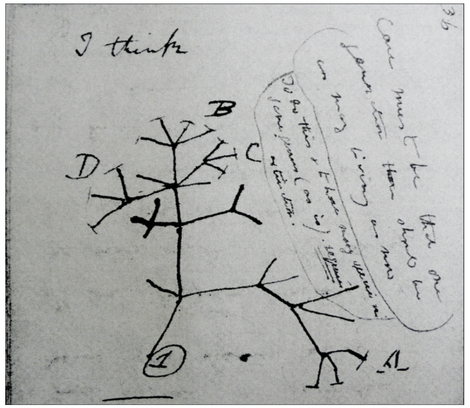


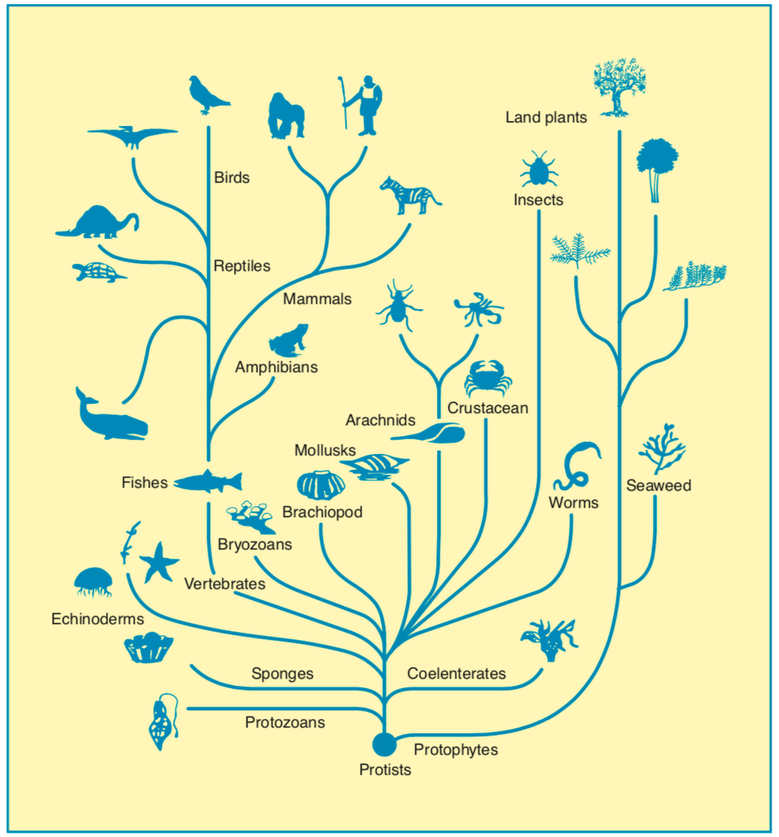
Fig. 2.1 Charles Darwin at about the time of meeting Romanes



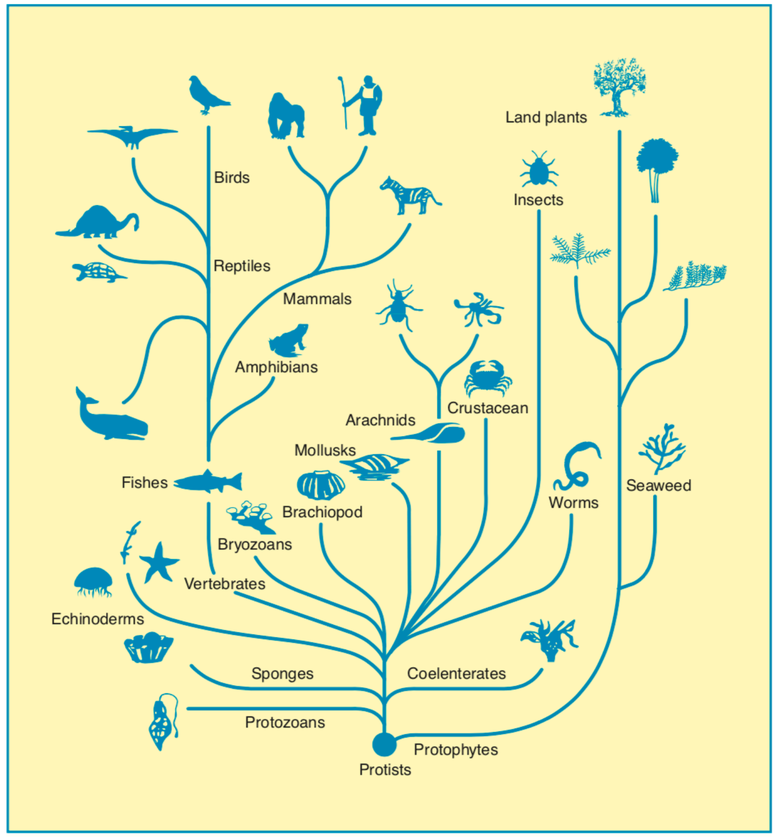
Insects → Birds → Apes → Humans

Evolution is not linear

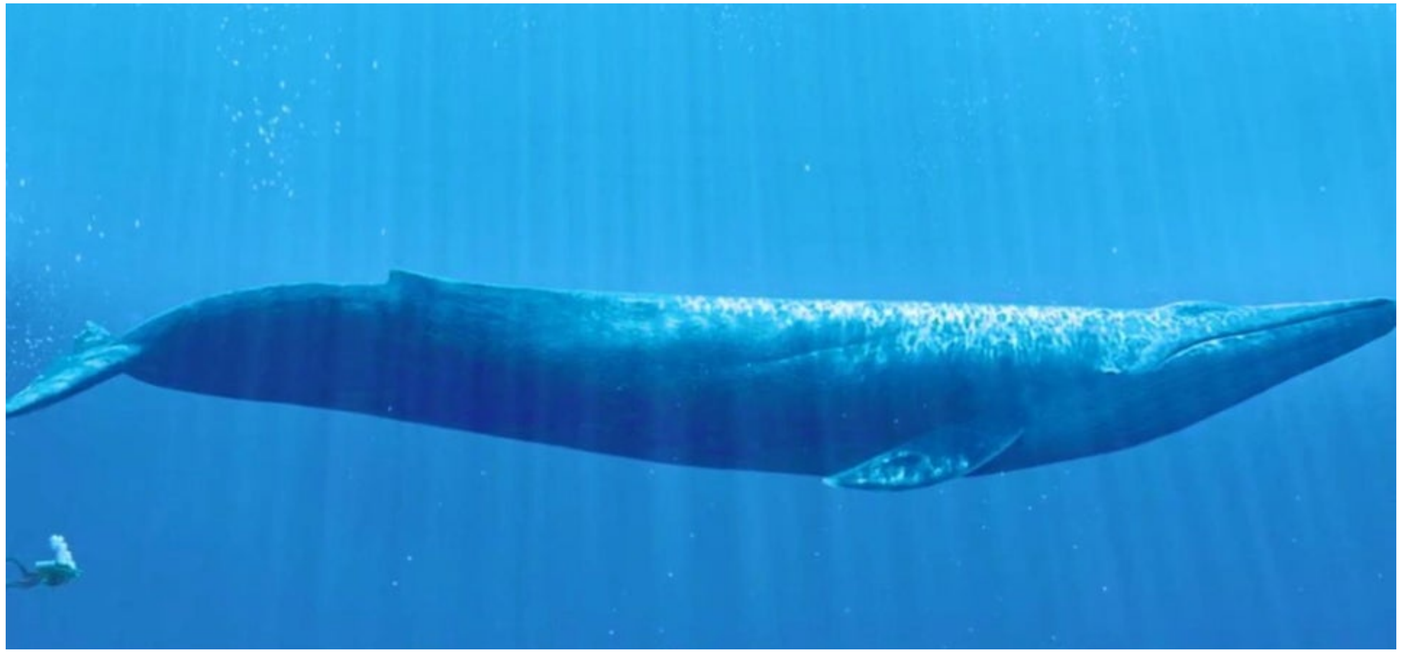




- Hinde and Stevenson-Hinde (1973) - multiple chances for intelligence to evolve
- Clayton and Emery (2005) - ‘feathered apes’ crow problem-solving

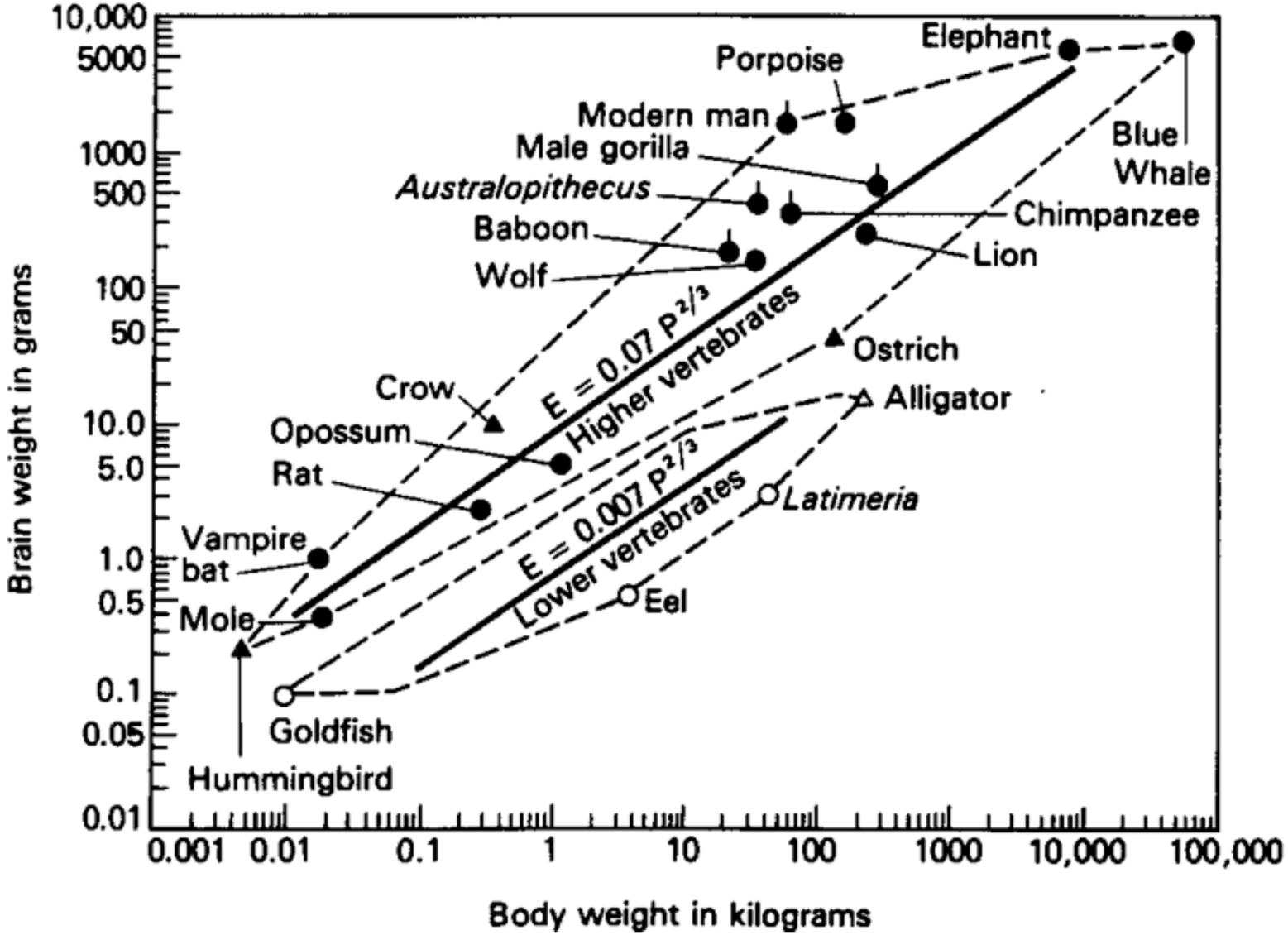


Brain size



Jerison (1973). The *principle of proper mass* - the more important a function is, the more of the brain is devoted to it

Brain size


Brain weight in grams
10,000
5000
1000
500
100
50
10.0
5.0
1.0
0.5
0.1
0.05
0.01
Body weight in kilograms
0.001
0.01
0.1
1
10
100
1000
10,000
100,000
Elephant
Porpoise
Modern man
Male gorilla
Australopithecus
Baboon
Wolf
Blue Whale
Chimpanzee
Lion
Ostrich
Alligator
Crow
Opossum
Rat
Vampire bat
Mole
Hummingbird
Goldfish
Eel
Latimeria
E = 0.07 P^2/3
Higher vertebrates
E = 0.007 P^2/3
Lower vertebrates

Brain size

Jerison (1973)

$$\frac{\text{Brain Weight (E)}}{\text{Body Weight (P)}}$$

$$\text{Cephalization Index, K} = \frac{E}{P^{2/3}}$$

Brain size

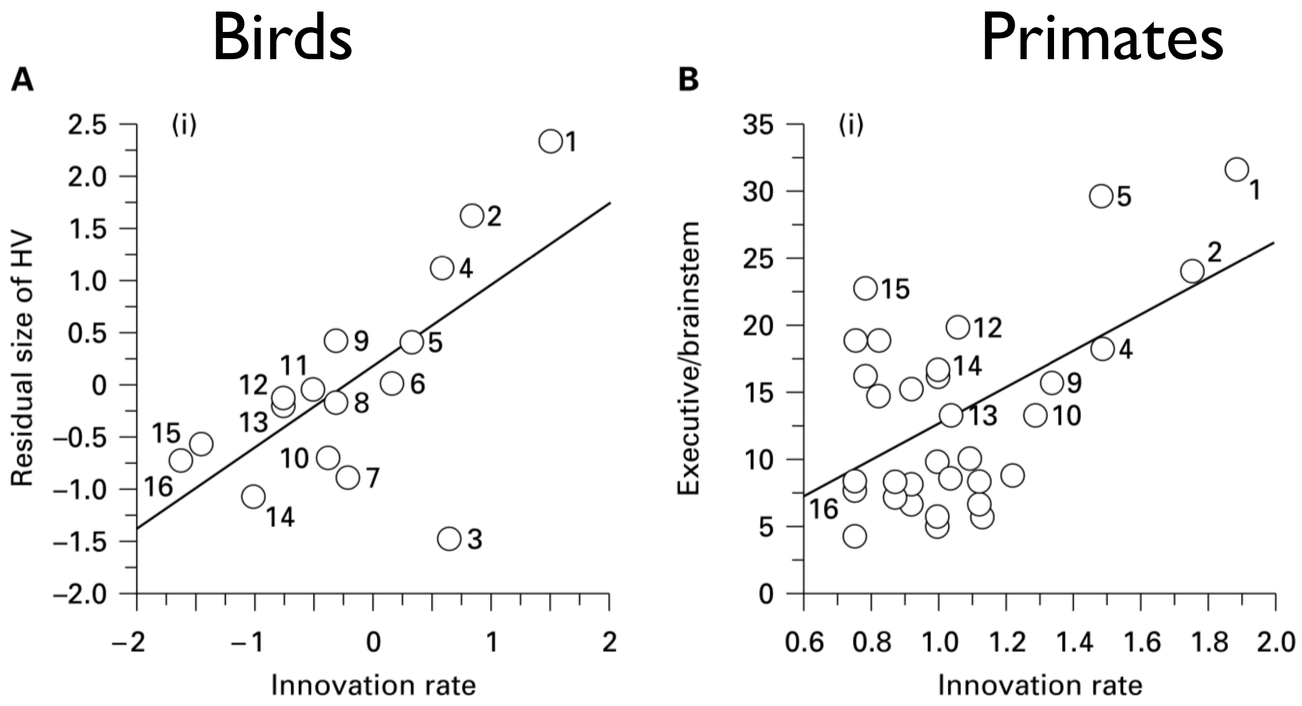
TABLE 1.2 The cephalization index (K) for 21 mammals arranged in descending order

Mammal	Score	Mammal	Score
Man	0.89	Dog	0.14
Dolphin	0.64	Squirrel	0.13
Chimpanzee	0.30	Wild pig	0.12
Squirrel monkey	0.28	Cat	0.12
Rhesus monkey	0.25	Horse	0.10
Elephant	0.22	Sheep	0.10
Whale	0.21	Ox	0.06
Marmoset	0.21	Mouse	0.06
Fox	0.19	Rat	0.05
Walrus	0.15	Rabbit	0.05
Camel	0.14		

Adapted from Russell (1979).

From Pearce (2008)

Brain size



- What processes are involved in 'innovation'?
- May relate more to 'boldness' than intelligence

Lefebvre, Reader, and Sol (2004)

Brain size

In addition

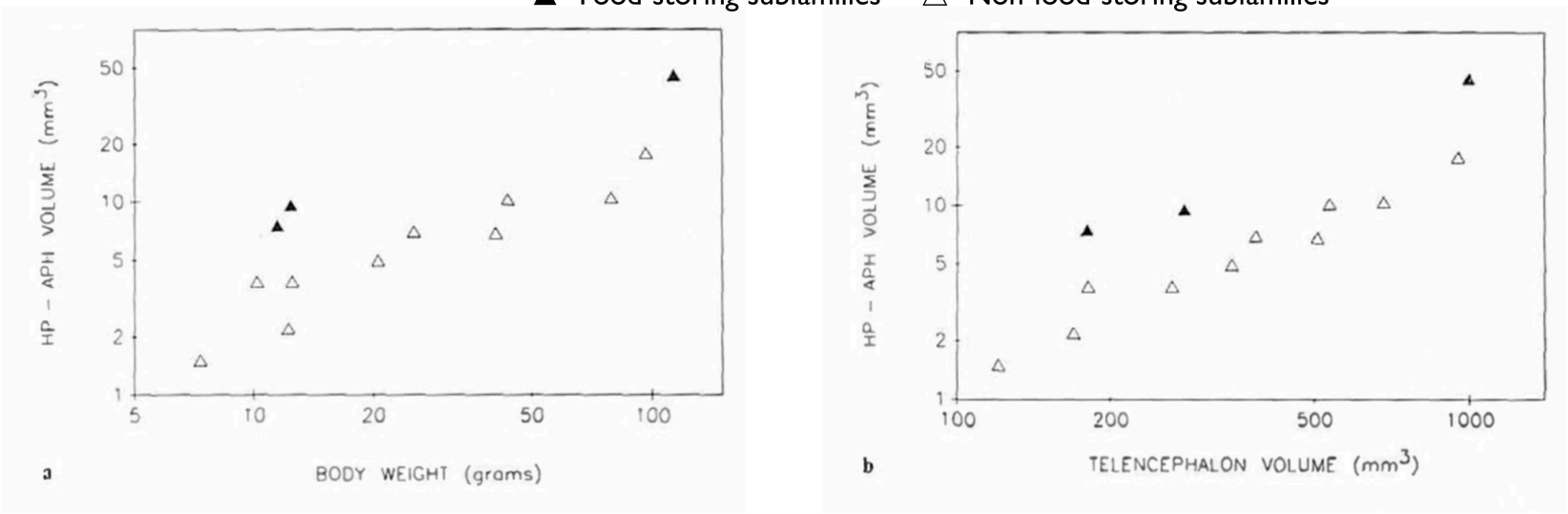
- Brain size has been argued to correlate with other 'intelligent' behaviours
  - Social intelligence hypothesis
  - Foraging intelligence hypothesis

- What processes are involved in 'innovation'?
- May relate more to 'boldness' than intelligence

Lefebvre, Reader, and Sol (2004)

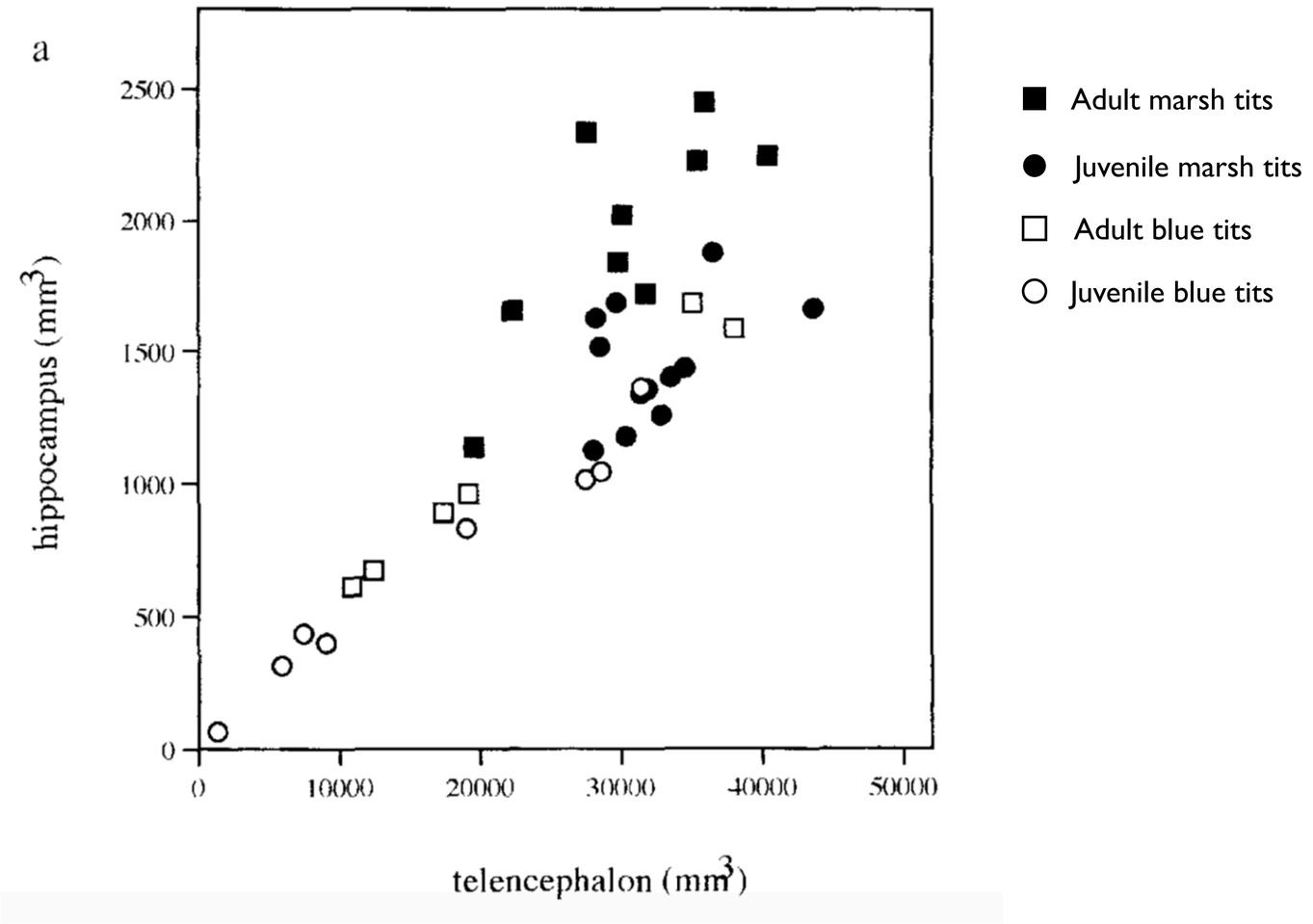
Brain size

▲ Food-storing subfamilies △ Non-food-storing subfamilies



Sherry et al. (1989)

Brain size

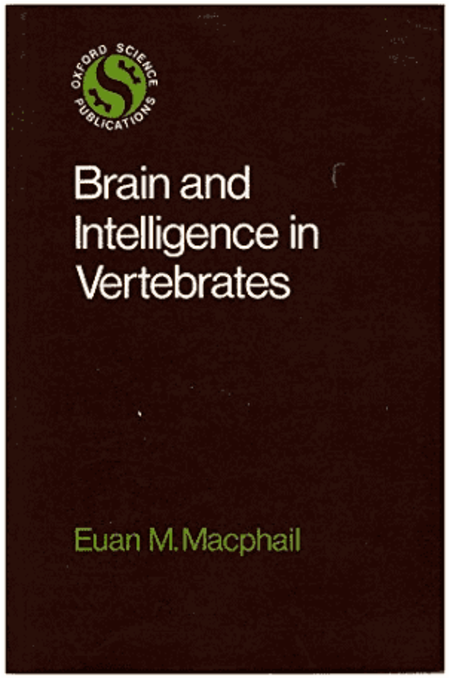


Healy, Clayton, and Krebs (1994)

Justifications for Rankings

- Appearance
- Aristotle's Great Chain of Being
- Evolution
- Brain Size

But see Macphail (1982)
- Null Hypothesis



Justifications for Rankings

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- Aristotle's Great Chain of Being
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But see Macphail (1982)
- Null Hypothesis

Learning

A relatively permanent change in behaviour resulting from experience

Operant or Instrumental conditioning

Response → Food

Pavlovian or Classical conditioning

Stimulus → Food

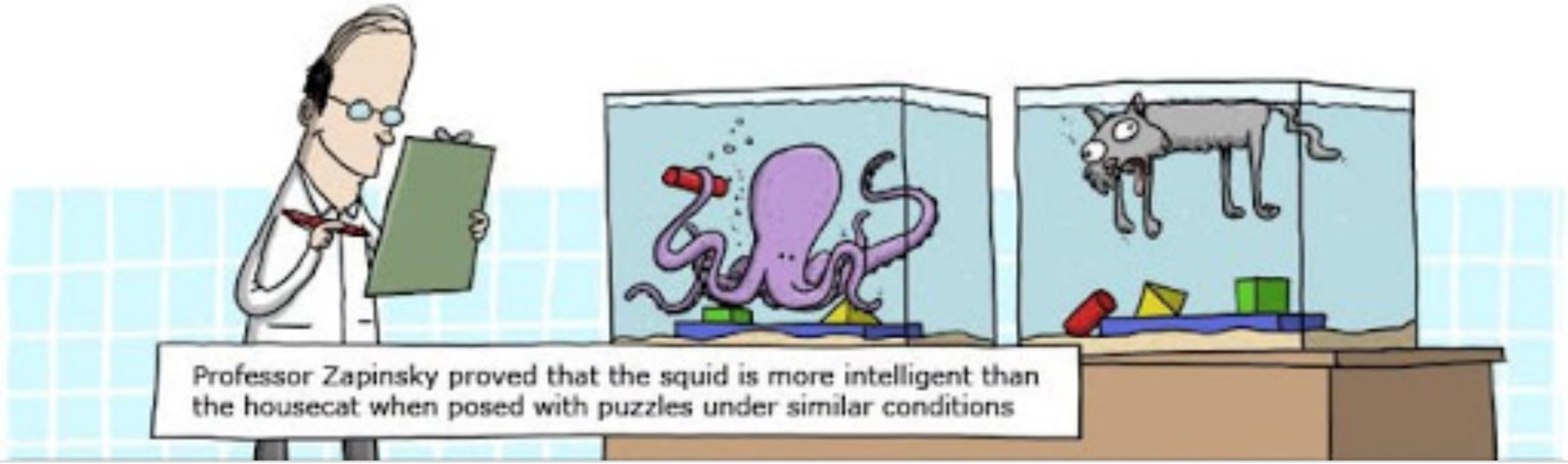
Problems with using speed of learning as a measure of intelligence

Unexpected between-species differences

- Skard (1950) - rats and humans in complex maze
- Warren (1965) - goldfish, chickens, cats, horses, monkeys in a stimulus discrimination
- Angermeier (1984) - operant response in various mammals, birds, fish
  - Papousek (1977) added 5 month-old children
  - Menzel & Erber (1978) added bees

Problems with using speed of learning as a measure of intelligence

Unexpected between-species differences



Problems with using speed of learning as a measure of intelligence

Unexpected between-species differences

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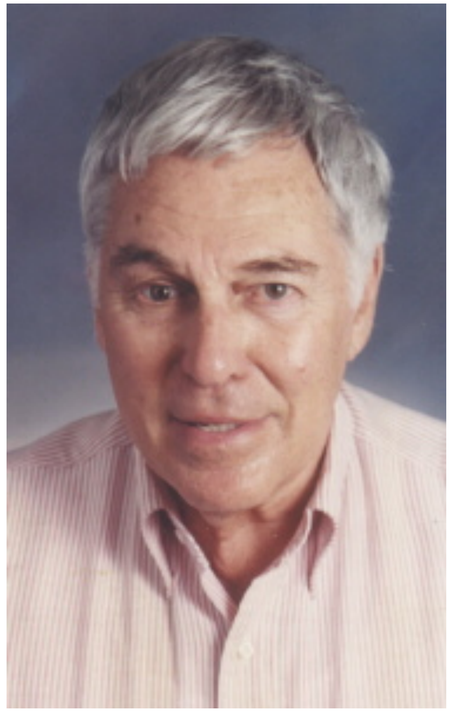
- Angermeier (1984) - operant response in various mammals, birds, fish
  - Papousek (1977) added 5 month-old children
  - Menzel & Erber (1978) added bees
  - Results showed the opposite of predictions from cephalisation index

Problems with using speed of learning as a measure of intelligence

Unexpected between-species differences



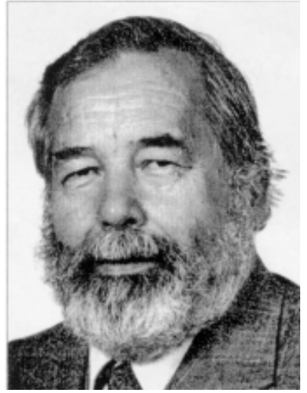
Why? Contextual variables

- Difficult to equate perceptual demands of test
- Difficult to equate motivational demands of test

Bitterman (1975) systematic variation

Problems with using speed of learning as a measure of intelligence

Within-species differences in speed of learning



Garcia & Koelling (1966)

Biological Relevance

Training	Test	Volume drunk
Saline → illness	Saline	
Saline → shock	Saline	
LTW → illness	LTW	
LTW → shock	LTW	

LTW = light + tone + water

So how do we measure animal intelligence?

What do we mean by animal intelligence?

“Unfortunately, there’s another elephant in the room.”

So how do we measure animal intelligence?

What do we mean by animal intelligence?

- Because ‘intelligence’ is conflated with the concept of a general IQ (in humans) we tend to avoid the term in relation to animals
- We tend to talk about ‘cognition’ instead
- Philosophers often equate ‘cognition’ and ‘reasoning’ (e.g., thinking, knowing, understanding)
- Psychologists tend to think of cognition as the ability to process information in an adaptive way
- Cognitive processes can be modelled in terms of computations

Animal Cognition as Information Processing

- Memory
- Learning
- Reasoning
- Navigation
- Communication